

R Programming Tutorial

#04

Hypothesis Testing

#lectureFIVE #01

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#04-1

INTERVAL ESTIMATION IN R

Interval Estimate in R for Proportions

REFER to:
lectureFIVE-ONE #01

EXAMPLE 2 (slide 25)

Example

- A recent study investigated the effects of a "Buckle Up Your Toddlers" campaign to get parents to use the grocery cart seat belts. Investigators observed a representative sample of parents at grocery stores in a large city, and found 192 out of 594 parents buckling up their toddlers.
 - a. Compute a point estimate of the true proportion of all parents who buckle up their toddlers.
 - b. Construct and interpret a 95% confidence interval for, the true proportion of all parents who buckle up their toddlers.

Interval Estimate in R for Proportions

WHAT WE KNOW?

- Sample size, n : 594
- 192 of the sample buckling up their toddlers.

WHAT WE WANT TO FIND?

- Point estimate of the true proportion.
- 95% confidence interval for the true proportion.

Example

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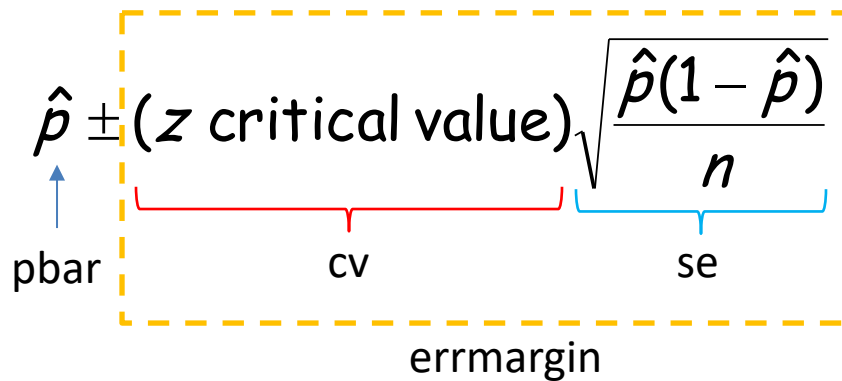
Interval Estimate in R for Proportions

WHAT WE KNOW?

- Sample size, n : 594
- 192 of the sample buckling up their toddlers.

WHAT WE WANT TO FIND?

- Point estimate of the true proportion.
- 95% confidence interval for the true proportion.



```

1      n = 594
2      k = 192
3      pbar = n/k;
4      se = sqrt(pbar*(1-pbar)/n)
5      cv = qnorm(0.975)
6      errmargin = se*cv;
7
8
9      // Get the interval
10     pbar+c(-errmargin,
11           errmargin)
12
13
14
15
16
17
18
19
20
21
22
  
```

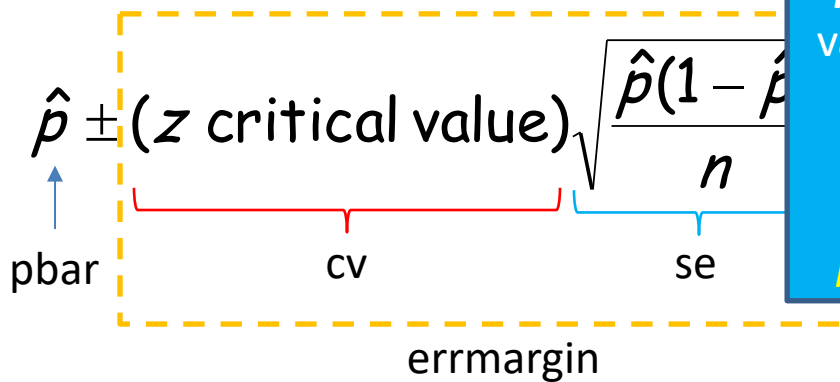
```
[1] 0.2856198 0.3608448
```

Interval Estimate in R for Proportions

WHAT WE KNOW?

$1 - \alpha$	α	$\alpha / 2$	$z_{\alpha/2}$
.90	.10	.05	$z_{.05} = 1.645$
.95	.05	.025	$z_{.025} = 1.96$
.98	.02	.01	$z_{.01} = 2.33$
.99	.01	.005	$z_{.005} = 2.575$

- 95% confidence interval for the true proportion.



```

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7
8
9      // Get the interval
10     pbar+c(-errmargin,
11            errmargin)

```

qnorm() can be used to get the Z critical value. E.g.: for 95% confidence level, $\alpha/2 = 0.025$, however, parameter to put in **qnorm()** is $1 - 0.025 = 0.975$

Ref: <http://seankross.com/notes/dpqr/>

```

20
21
22

```

```
[1] 0.2856198 0.3608448
```

Interval Estimate in R for Proportions

Solution

a) point estimate: $p = 192 / 594 = 0.3232$

b) 95% CI: = 95% CI: $0.3232 \pm 1.960 \sqrt{\frac{0.3232(0.6768)}{594}} = (0.2856, 0.3608)$.

I am 95% confident that the true proportion of parents who buckle up their toddlers is between 0.286 and 0.361.

```
[1] 0.2856198 0.3608448
```

Interval Estimate in R for Proportions (Alternative)

a) point estimate: $p = 192 / 594 = 0.3232$

b) 95% CI: = 95% CI: $0.3232 \pm 1.960 \sqrt{\frac{0.3232(0.6768)}{594}} = (0.2856, 0.3608)$.

I am 95% confident that the true proportion of parents who buckle up their toddlers is between 0.286 and 0.361.

1-sample proportions test with continuity correction

```
data: k out of n, null probability 0.5
X-squared = 73.537, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.2860486 0.3627384
sample estimates:
      p
0.3232323
```

Interval Estimate in R for Proportions (Alternative)

`prop.test()` is used to test proportions.
 It includes the calculation of confidence interval
 (95% by default)

How to use `prop.test()`:

```
prop.test(x, n, p = NULL,
          alternative = c("two.sided", "less", "greater"),
          conf.level = 0.95, correct = TRUE)
```

Ref: <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/prop.test.html>

```
1      n = 594
2      k = 192
3
4      prop.test(k, n)
5
6
7
8
9
10
11
```

1-sample proportions test with continuity
 correction

```
data:  k out of n, null probability 0.5
X-squared = 73.537, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.2860486 0.3627384
```

sample estimates:

```
      p
0.3232323
```

Confidence Interval (Default: 95%)

#04-2

INTERVAL ESTIMATION FOR MEAN (VARIANCE KNOWN)

Interval Estimate in R for Mean

REFER to:
lectureFIVE-ONE #02

EXAMPLE 2 (slide 30)

Example

- A computer company samples demand during lead time over 25 time periods:

235	374	309	499	253
421	361	514	462	369
394	439	348	344	330
261	374	302	466	535
386	316	296	332	334

It is known that the standard deviation of demand over lead time is 75 computers. We want to estimate the **mean** demand over lead time with 95% confidence in order to set inventory levels

Interval Estimate in R for Mean

WHAT WE KNOW?

- Sample size, n : 25
- Standard deviation: 75

WHAT WE WANT TO FIND?

- 95% confidence interval for the mean.

Example

- A computer company samples demand during lead time over 25 time periods:

235	374	309	499	253
421	361	514	462	369
394	439	348	344	330
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It is known that the standard deviation of demand over lead time is 75 computers. We want to estimate the **mean** demand over lead time with 95% confidence in order to set inventory levels

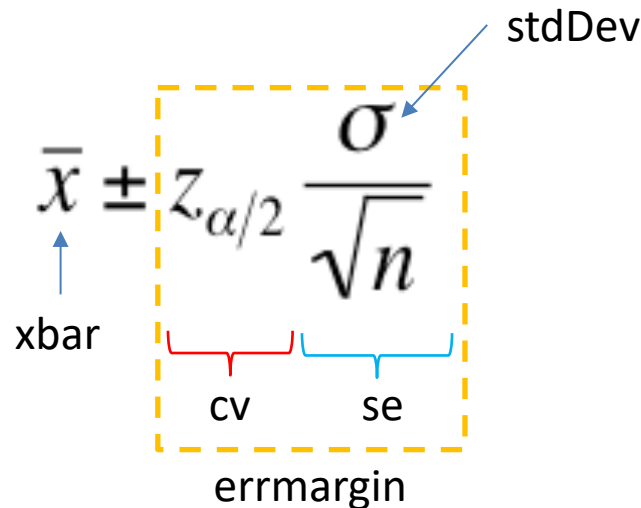
Interval Estimate in R for Mean

WHAT WE KNOW?

- Sample size, n: 25
- Standard deviation: 75

WHAT WE WANT TO FIND?

- 95% confidence interval for the mean.



$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Diagram labels: $\bar{x}$ (xbar), $z_{\alpha/2}$ (cv), $\frac{\sigma}{\sqrt{n}}$ (se), σ (stdDev), and the entire expression is enclosed in a dashed box labeled **errmargin**.

```

1      n = 25
2      stdDev = 75
3      x = c(235, 374, 309, 499, 253,
4            421, 361, 514, 462, 369,
5            394, 439, 348, 344, 330,
6            261, 374, 302, 466, 535,
7
8            386, 316, 296, 332, 334)
9      xbar = mean(x)
10     se = stdDev/sqrt(n)
11     cv = qnorm(0.975)
12     errmargin = cv*se
13
14     // 95% C.I.
15     xbar+c(-errmargin, errmargin)
16
17
18
19
20
21
22
  
```

```
[1] 340.7605 399.5595
```

Interval Estimate in R for Mean

Solution

- In order to use our confidence interval estimator, we need the following pieces of data:

$\bar{x}$	370.16	} Calculated from the data...
$z_{\alpha/2}$	1.96	
σ	75	} Given
n	25	

$1 - \alpha = .95, \therefore \alpha/2 = .025$
 so $z_{\alpha/2} = z_{0.025} = 1.96$

therefore: $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 370.16 \pm 1.96 \frac{75}{\sqrt{25}} = 370.16 \pm 1.96 \frac{75}{5} = 370.16 \pm 29.40$

The **lower** and **upper** confidence limits are **340.76** and **399.56**.

```

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2      stdDev = 75
3      x = c(235, 374, 309, 499, 253,
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7            386, 316, 296, 332, 334)
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